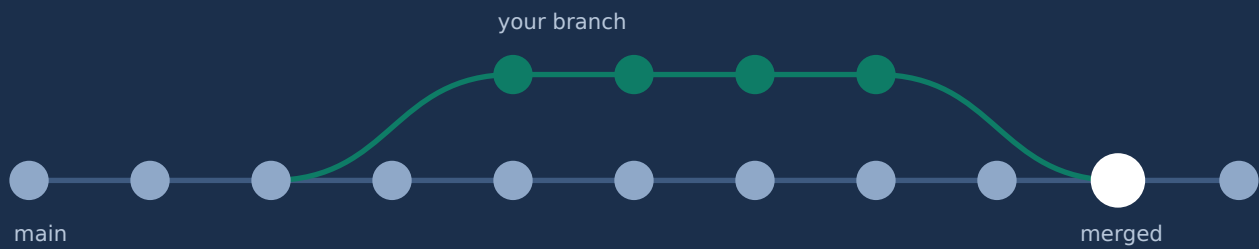


EDITION: OCTOBER 2026



Open Source Contribution Handbook

From your first pull request to GSoC and LFX Mentorship

Programs and stipends for Indian students · 100 organisations in 3 tiers
Step-by-step workflow with AI code editors · AI policies you must follow
Proposal guide · 12-month plan · Events calendar Oct 2026 - Nov 2027

For 2nd and 3rd year Computer Science students

Companion workbook: [Open_Source_Contribution_Guide_Oct2026.xlsx](#)

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How to read this book

- Chapters 1-4 explain the programs. Read them once.
- Chapters 5-10 are the practical part: skills, choosing an org, the PR workflow, AI rules and CI checks. Keep them open while you work.
- Chapters 11-13 help you apply and plan your year. Chapters 14-15 are reference tables.
- Every table here is also in the Excel workbook, where you can filter, sort and track your progress.
- Facts were checked on official sites on 1 October 2026. Dates marked 'expected' are projections from past years; always confirm on the official link.

CHAPTER 1

Why open source?

Open source is the fastest way for a student to work on real software, get reviewed by experienced engineers, and build public proof of skill.

Open source software is code that anyone can read, use and improve. Linux, Kubernetes, Python, VS Code, React and most of the tools you use every day are built this way, by thousands of contributors from companies, universities and their own homes.

When your pull request (PR) is merged into a real project, three things happen. Your code runs for real users. An experienced maintainer has reviewed your work line by line, which is the best free mentoring you can get. And your GitHub profile now shows a public, verifiable record that recruiters and program mentors can check.

What you get out of it

- **Skills that college projects don't teach:** reading a large codebase, writing tests, following style guides, passing CI, and handling code review.
- **Stipends:** GSoC pays Indian contributors \$750-\$3,000 depending on project size, LFX pays \$1,300, and C4GT's DMP pays Rs 1,00,000.
- **A network:** maintainers often work at companies like Google, Red Hat, Microsoft, Intel and many startups. Good contributors get referrals.
- **A portfolio:** merged PRs in known projects make a far stronger resume line than a to-do app.
- **Confidence:** after your first few merges, you stop being afraid of big codebases.

What counts as a contribution

Code is only one part. Maintainers value all of these, and several are good first steps:

- Fixing bugs and adding small features
- Improving documentation, READMEs and tutorials
- Writing or improving tests
- Reproducing and triaging bug reports
- Reviewing other people's PRs (politely)
- UI/UX design, translations and accessibility fixes
- Answering questions from other newcomers in the community chat

The right mindset

You are a guest in someone else's project. Maintainers are busy, and many are volunteers. The contributors who succeed are the ones who make a maintainer's life easier: they read the docs before asking, keep PRs small, test their changes, and reply to review comments quickly and politely. One careful merged PR is worth more than ten careless ones.

Key idea

- Quality over quantity. Consistency over months beats a burst of activity in March.
- Pick one main organisation and grow there. Depth is what mentors notice.

CHAPTER 2

What changed in 2026

Several rules changed this year. Many blog posts and lists online are now out of date. Read this chapter before you plan anything.

Change	What it means for you	Source
LFX stipend model changed (1 Jul 2026)	Stipend is now based on purchasing power per country. India = \$1,300 total for programs starting on or after 1 Jul 2026 (worldwide range \$1,000-\$6,600).	LF docs
GSoC stipend for India	\$750 small (~90 h), \$1,500 medium (~175 h), \$3,000 large (~350 h).	GSoC stipends
GSoC AI guidance published	Some orgs ban AI completely, even for proposals. AI-written proposals can be rejected automatically.	GSoC AI guidance
Hacktoberfest 2026: PRs no longer count	Rewards come from events, livestreams and challenges. Don't send random PRs for Hacktoberfest.	Hacktoberfest FAQ
MLH Fellowship not open to India	The official FAQ says the program is not open to Asia-Pacific countries right now.	MLH Fellowship
Outreachy rules for Indian students	Indian university students can apply only for the May-Aug cohort. Past GSoC contributors are not eligible.	Outreachy docs
C4GT renamed	Code for GovTech is now Code for Good Tech, run from IIIT Hyderabad. DMP still pays Rs 1 lakh.	IIIT Hyderabad blog
Programs that ended	Google Season of Docs (ended 2024), X.Org EVoC (sunset), FOSSASIA Codeheat (inactive since 2024), Google Code-in (cancelled 2020).	Official sites
AI rules everywhere	Many repos now have AI_POLICY.md or AGENTS.md. Some (QEMU, Gentoo, NetBSD, GCC) ban AI-generated code.	Chapter 8
GitHub PR limits	Maintainers can now limit open PRs per new contributor (Jun 2026). Jaeger allows first-timers only 1 open PR.	GitHub changelog
Free AI tool tiers changed	Gemini Code Assist's free tier and Gemini CLI's free Google login ended on 18 Jun 2026 (move to Antigravity). GitHub moved students to a new 'Copilot Student' plan.	Chapter 8

Corrections to common lists

Popular lists (including some that were shared with this book) contain entries that are no longer accurate:

- **'GNOME Summer of Code'** is not a separate program; GNOME takes part through GSoC.
- **Google Season of Docs** is not running in 2026 or 2027.
- **MLH Fellowship** is listed as open to all, but India and APAC are currently excluded.
- **LFX '\$3,000-\$6,600'** is the old range. For India it is now \$1,300.
- **Hacktoberfest '4 PRs'** is the old rule. In 2026 PRs don't count at all.

CHAPTER 3

The program landscape

There are three kinds of programs: paid mentorships, unpaid mentorships, and contribution drives. Use unpaid ones to build experience; aim paid ones at the right time.

Paid programs open to Indian students

Program	Stipend (India)	Duration	Next window
Google Summer of Code (GSoC)	India: \$750 (small, ~90 h) / \$1,500 (medium, ~175 h) / \$3,000 (large, ~350 h). Paid 45% after midterm, 55% after final.	Standard 12 weeks; can be extended up to 22 weeks	2027 dates not announced. Expected: orgs announced ~late Feb 2027, contributor applications ~mid-late Mar 2027, coding ~late May 2027.
LFX Mentorship (incl. CNCF)	India: \$1,300 total (new PPP-based model for programs starting on/after 1 Jul 2026; range \$1,000-\$6,600 worldwide). Paid in 2 parts.	12 weeks full-time or 24 weeks part-time	2027 Term 1 expected: projects listed ~mid-Jan, mentee applications ~late Jan-mid Feb 2027, program Mar-May 2027.
Linux Kernel Mentorship Program (LKMP)	Location-based where funded. Some sessions are unpaid (credit only).	12 weeks full-time or 24 weeks part-time	Spring 2027 expected: listings ~mid-Jan 2027, start ~Mar 2027.
LF Decentralized Trust (Hyperledger) Mentorship	Tiered by country of residence; India amount not published for 2026.	About 24 weeks part-time (Jun-Nov)	Expected mentee applications ~Apr-May 2027.
Open Mainframe Project Mentorship	Not stated on the 2026 page.	About 3 months (Jun-Aug)	Expected applications ~Apr-May 2027.
LF Networking (LFN) Mentorship	LF stipend policy for funded projects.	12 weeks FT or 24 weeks PT	Follows the LF term calendar (Jan / Apr / Jul listings).
Outreachy	\$7,000 total.	13 weeks, 30 h/week	May 2027 cohort expected: initial applications ~early-mid Feb 2027, contribution period ~mid-Mar to mid-Apr.
C4GT Dedicated Mentoring Program (DMP)	Rs 1,00,000	About 3 months	Expected DMP 2027 applications ~Apr-May 2027.
Summer of Bitcoin	\$3,000-\$6,600 equivalent paid in BTC, PPP-based (India amount not listed).	12 weeks	Expected applications ~Jan-mid Feb 2027.
European Summer of Code (ESoC)	EUR 4,800 for 3 months full-time equivalent (pro-rata).	12 weeks	Expected applications from ~mid-Feb 2027.
Open Source Promotion Plan (OSPP, China)	CNY 3,500-10,000 depending on project mode (2026).	Project-based	Expected launch ~Mar-Apr 2027.
Igalia Coding Experience	EUR 7,000 for 450 hours.	3 or 6 months	Expected call ~Feb-Apr 2027.
Ethereum Protocol Fellowship (EPF)	Monthly stipend for selected participants (amount not published).	About 5 months	Expected cohort 8 applications ~Apr-May 2027.
Processing Foundation microgrants	\$500 for ~10 hours of p5.js work	Short	Check the grants page.

Unpaid mentorships and contribution drives

Program	What you get	When	Link
FOSSEE Summer Fellowship (IIT Bombay)	Stipend not confirmed for 2026 (certificate).	Expected screening tasks ~Mar-Apr 2027.	fossee.in/fellowship/2026
Season of KDE	None	Expected applications ~mid-Dec 2026 to mid-Jan 2027.	mentorship.kde.org/sok

Program	What you get	When	Link
GirlScript Summer of Code (GSSoC)	None	Expected announcement ~Jan 2027.	gssoc.girlscript.org
Social Winter of Code (SWoC) Season 7	None	REGISTRATION OPEN NOW (till Dec 31, 2026); coding Jan 1-Mar 30, 2027.	www.swoc.in
Kharagpur Winter of Code (KWoC)	None	Expected registration ~late Nov 2026; coding Dec 2026-Jan 2027.	kwoc.kossiitkgp.org
Hacktoberfest 2026	Stickers (3 virtual stickers = physical sticker pack)	RUNNING NOW (October 2026).	hacktoberfest.com
24 Pull Requests	None	Dec 1-24, 2026.	24pullrequests.com
Djangonaut Space	None	Expected Session 8 applications ~Jan 2027.	djangonaut.space
Kubernetes Release Team Shadow	None	v1.39 shadow applications expected ~Dec 2026.	github.com/kubernetes/sig-release/blob/master/release-team/shadows.md
Swift Mentorship Program	None	Expected survey ~late May-Jun 2027.	www.swift.org/mentorship
KubeStellar IFoS (Interns for Open Source)	None (certificate + recommendation letter)	Rolling (apply any time).	kubestellar.io/en/programs/ifos
Julia Seasons of Contributions (JSoC)	Same as GSoC	Follows GSoC 2027.	julialang.org/jsoc
Summer of Haskell	Same as GSoC	Hopes to join GSoC 2027.	summer.haskell.org
sktime mentoring	None (paid only via GSoC/ESoC)	Rolling.	github.com/sktime/mentoring
Nexus Spring of Code (NSoC)	Prize pool (unverified)	Expected ~Mar-May 2027 (unverified).	www.nsoc.in
Apertre	Prizes for top contributors	Check the site.	apertre.resourcio.in

Not available or discontinued

Program	Status	Detail
MLH Fellowship	Not available for India	NOT currently open to India or any Asia-Pacific country (official FAQ).
Google Season of Docs	Discontinued	Ended after 2024.
X.Org EVoC	Discontinued	Sunset; not accepting applications.
FOSSASIA Codeheat	Inactive	Last ran Dec 2023-Mar 2024.
Google Code-in	Discontinued	Cancelled in 2020.
FSF Internships	No current call	Last posted deadline Nov 2023.

Full details (eligibility, 2026 dates, how to apply) for all 36 programs are in the **Programs** sheet of the workbook.

CHAPTER 4

GSoC and LFX in detail

These two are the main targets for most Indian students. They work differently, so prepare for them differently.

Google Summer of Code (GSoC)

Google pays contributors to work on an open-source project under mentors from an approved organisation. The standard project is 12 weeks and can be extended up to 22 weeks. In 2026, 185 organisations were announced (184 took part). 15,245 people from 131 countries sent 23,371 proposals, and 1,141 contributors were accepted, which is about 1 in 13 applicants.

Item	Details
Who can apply	18+. Students, or people who are beginners to open source (see the FAQ). You must not have been accepted into GSoC more than once before, so you can take part at most twice.
How many proposals	Up to 3 proposals; only 1 can be accepted.
Project sizes	Small (~90 hours), Medium (~175 hours), Large (~350 hours).
Stipend (India)	\$750 / \$1,500 / \$3,000. Paid 45% after the midterm and 55% after the final evaluation.
2026 timeline (reference)	Orgs announced Feb 19. Contributor applications Mar 16-31. Results Apr 30. Community bonding May 1-24. Coding from May 25. Midterms Jul 6-10. Standard end Aug 24-31; extended projects until Nov 2.
2027 (expected)	No 2027 timeline yet. Expect orgs ~late Feb 2027 and applications ~mid-late March 2027.
What orgs look for	Merged contributions in their repo, clear communication, a realistic proposal written by you, and evidence you understand their code.

How GSoC selection really works

1. Organisations publish an **ideas list**. You can also propose your own idea, but discuss it with mentors first.
2. Before applications open, contributors join the org's chat, fix small issues and discuss ideas with mentors.
3. Many orgs require a **pre-task** or a minimum number of merged PRs. Check the org's GSoC page.
4. You submit the proposal on the GSoC site. Mentors rank proposals. Google allots slots to each org.
5. Your contribution history and the quality of your discussions usually matter as much as the proposal document.

LFX Mentorship (Linux Foundation, incl. CNCF)

LFX runs three 12-week terms every year for Linux Foundation projects. Most projects come from the CNCF (Kubernetes ecosystem), but LF Decentralized Trust, the Linux kernel (LKMP), Open Mainframe and LF Networking also use it.

Item	Details
Terms	Term 1: Mar-May. Term 2: Jun-Aug. Term 3: Sep-Nov. Part-time (24 weeks) is possible for some projects.
Who can apply	18+. You can be selected for an LF mentorship only once in your life, so choose well.
Applications	Max 3 applications per term. Resume + cover letter, and often a small task or test.
Stipend (India)	\$1,300 total, in two instalments (after midterm and final).

Item	Details
Where projects are listed	github.com/cncf/mentoring (CNCF projects by term) and mentorship.lfx.linuxfoundation.org .
2026 Term 3 (reference)	Proposals Jul 1-28, mentee applications Aug 3-18, program Sep 7-Nov 27.
2027 Term 1 (expected)	Projects listed ~mid-January; applications ~late Jan to mid-Feb 2027; program March-May.

GSoC vs LFX: how to prepare

	GSoC	LFX
Application	Long proposal with timeline	Resume + short cover letter (+ task)
Main signal	Proposal + contributions	Contributions + fit for the specific project
Frequency	Once a year	Three terms a year
Limit	Accepted max twice ever	Selected once ever (all LF programs)
Stipend (India)	\$750-\$3,000	\$1,300
Best strategy	Contribute to one org from Oct-Mar, then write the proposal with mentors	Contribute to a CNCF project; apply the moment its project is listed

CHAPTER 5

Skills and learning paths

You don't need to be an expert. You need Git, one stack you are comfortable with, and the patience to read other people's code.

Skills everyone needs

- **Git and GitHub:** clone, branch, commit, rebase, push, pull requests, resolving conflicts. See Appendix A.
- **Linux command line:** navigating folders, running scripts, reading logs, installing packages.
- **Reading code:** finding where something happens, following function calls, using search (Ctrl+Shift+F, grep, git grep).
- **Testing:** running a project's test suite and writing a small test for your change.
- **Written English:** clear, short messages. Nobody expects perfect grammar, but they expect clarity.

Pick one track

Track	Learn first	Good orgs to start (tier)
Web (JS/TS/React/Angular/Vue)	Modern JS, TypeScript, React basics, npm, Jest/Vitest	Jaeger UI, Open Healthcare Network, CircuitVerse, Kiwix, p5.js, Oppia, OpenMRS (Beginner); Headlamp, OpenEverest (Intermediate)
Python / Django	Python, virtualenv, pytest, Django basics	Zulip, Open Library, Wagtail, SymPy, Kolibri (Beginner); Django, Kubeflow, krkn, OpenWISP (Intermediate)
Go / Cloud native	Go basics, Docker, Kubernetes with kind or minikube	Meshery (Beginner); Kyverno, Kubescape, PipeCD, KubeEdge, Harbor, Lima (Intermediate)
Mobile (Android/Flutter)	Kotlin or Dart/Flutter	AnkiDroid, API Dash, Open Food Facts (Beginner)
Java	Java, Maven/Gradle, JUnit	JabRef (Beginner); Jenkins, checkstyle, Apicurio, Microcks (Intermediate)
Systems (C/C++/Rust)	C, pointers, Makefiles, debugging with gdb; Rust basics	Graphite (Intermediate); LLVM, Git, Linux kernel via LFD103, FFmpeg (Advanced)
AI / ML	Python, NumPy, PyTorch	Kornia, OpenVINO, Kubeflow (Intermediate)

Suggested order for a 2nd year student

1. Weeks 1-2: Git and GitHub basics. Make a practice PR on freeCodeCamp or first-contributions style repos.
2. Weeks 3-4: Pick a Beginner org in your stack. Set it up locally and fix a documentation or small UI issue.
3. Months 2-4: 3-5 merged PRs in the same org. Start reading how its main features work.
4. Months 5-6: Take a medium-size issue. Talk to mentors about GSoC/LFX ideas.
5. 3rd year: move to an Intermediate org if you want cloud-native or systems experience.

CHAPTER 6

Choosing the right organisation

The org you choose decides most of your experience. Pick an active, friendly project that uses a stack you like.

Checklist before you commit

Check	How to check	Good sign
Is it active?	Commits in the last 30 days; Insights > Pulse on GitHub	Several merges per week
Do outsiders get merged?	Closed PRs from people who are not maintainers	Newcomer PRs merged within 1-2 weeks
Do maintainers reply?	Look at recent issues and PR comments	Replies within a few days
Are there starter issues?	Labels: good first issue, help wanted, easy	Unassigned issues from the last 3 months
Is setup documented?	CONTRIBUTING.md, dev-setup docs	You can run it locally in an evening
Is there a community?	Slack, Discord, Zulip, forum, community calls	Newcomers get answers
Is it in a program?	GSoC org list, cncf/mentoring, LFX site	Took part in 2025 or 2026
Do you care about it?	Would you use it?	You'll stay motivated for months

Red flags

- Last commit months ago, or dozens of open PRs with no review.
- Rude or dismissive replies to newcomers.
- No CONTRIBUTING guide and no chat channel.
- Every good-first issue already has 5 people asking for it (move to recent bugs instead).

The tiers used in this book

Tier	What it means	Who it suits
Beginner	Web/Python/mobile stacks, strong newcomer docs, many small issues, quick setup.	2nd year, first org
Intermediate	Cloud-native (Go/Kubernetes), ML, bigger codebases, strict CI (DCO, conventional commits).	3rd year, or your second org
Advanced	Systems work (C/C++/Rust), compilers, kernels, multimedia, science. Often mailing-list patches.	Strong fundamentals, lots of patience

Top picks by tier

Organisation	Tier	Programs 2026	Stack	Starter repo	Commits Sep
Jaeger (CNCF)	Beginner	GSoC 2026 (via CNCF) + LFX 2026 (T1, T2, T3)	React, TypeScript	jaegertracing/jaeger-ui	73
Oppia Foundation	Beginner	GSoC 2026	Angular, TypeScript, Python	oppia/oppia	61
Zulip	Beginner	GSoC 2026	Python/Django, TypeScript	zulip/zulip	273
Meshery	Beginner	GSoC 2026 + LFX 2026 (T1, T2, T3)	React/Next.js, Go	meshery/meshery	317

Organisation	Tier	Programs 2026	Stack	Starter repo	Commits Sep
Open Healthcare Network	Beginner	GSoC 2026	React, TypeScript	ohcnetwork/care_fe	38
Internet Archive	Beginner	GSoC 2026	Python, JavaScript	internetarchive/openlibrary	145
Headlamp (Kubernetes SIG UI)	Intermediate	LFX 2026 (T3)	React, TypeScript, Go	kubernetes-sigs/headlamp	92
Kyverno (Nirmata)	Intermediate	LFX 2026 (T1, T2, T3)	Go	kyverno/kyverno	87
Kubescape (ARMO)	Intermediate	LFX 2026 (T2, T3)	Go	kubescape/kubescape	177
PipeCD	Intermediate	GSoC 2026 (via CNCF) + LFX 2026 (T1, T2, T3)	Go, TypeScript	pipe-cd/pipecd	91
Kubeflow	Intermediate	GSoC 2026 + LFX 2026 (T3)	Python, Go, TypeScript	kubeflow/pipelines	155
Jenkins	Intermediate	GSoC 2026	Java, JavaScript	jenkinsci/jenkins	98
Kubernetes	Advanced	LFX 2026 (T1, T2, T3)	Go	kubernetes/kubernetes	700
LLVM	Advanced	GSoC 2026	C++	llvm/llvm-project	4724
Linux Kernel (LKMP)	Advanced	LF Kernel Mentorship (LKMP) + GSoC 2026 (The Linux Foundation)	C	torvalds/linux	1895
WasmEdge	Advanced	GSoC 2026 (via CNCF) + LFX 2026 (T1, T2, T3)	C++, Rust	WasmEdge/WasmEdge	66
Rust Foundation	Advanced	GSoC 2026	Rust	rust-lang/rust	2172
Git	Advanced	GSoC 2026	C	git/git	110

The full list of 100 organisations is in Chapter 15 and in the **100 Orgs** sheet, with links to good-first issues, contributing guides and AI rules.

CHAPTER 7

The contribution workflow, step by step

From fork to merge. This is the loop you will repeat for every contribution, with AI helping where it is allowed.

The loop in one line

- Read rules -> claim issue -> fork -> clone -> branch -> understand -> change -> test -> commit -> push -> PR -> CI -> review -> merge -> repeat.

Step 1. Set up

Create a GitHub account with a real name and photo, set up Git, SSH key and 2FA. Apply for the GitHub Student Developer Pack.

```
git config --global user.name "Your Name"
git config --global user.email you@example.com
ssh-keygen -t ed25519 -C you@example.com
```

- **AI:** Ask AI to explain any Git error message.
- **Watch out:** Use the same email in Git and GitHub, or commits won't link to your profile.

Step 2. Choose an org

Pick 2-3 orgs from the '100 Orgs' sheet that match your stack. Check activity, good-first issues and chat channel.

- **AI:** Ask AI to summarise the org's README and ideas page.
- **Watch out:** Don't spread across 10 orgs at once. Depth beats breadth.

Step 3. Read the rules

Read CONTRIBUTING, CODE_OF_CONDUCT, the PR template and any AI_POLICY.md / AGENTS.md. Note DCO, CLA and commit-message rules.

- **AI:** Ask AI: 'Turn CONTRIBUTING.md into a checklist for me.'
- **Watch out:** If AI is banned in that project, don't use it for code there.

Step 4. Join the community

Join the Slack/Discord/Zulip/mailling list. Introduce yourself in your own words. Attend a community call.

- **AI:** None. Write your own messages.
- **Watch out:** Many orgs ban AI-written chat messages.

Step 5. Find and claim an issue

Filter 'good first issue' or recent bugs. Comment to ask for assignment; wait for a maintainer reply before coding.

```
Label filter: is:issue is:open label:"good first issue" no:assignee
```

- **AI:** Ask AI which files are probably related to the issue.
- **Watch out:** Never take an issue already assigned to someone else.

Step 6. Fork

Click Fork on GitHub to copy the repo to your account.

```
(GitHub UI) Fork
```

- **Watch out:** Keep the fork's default branch clean; work in branches.

Step 7. Clone + upstream

Clone your fork and add the original repo as 'upstream'.

```
git clone git@github.com:<you>/<repo>.git
cd <repo>
git remote add upstream https://github.com/<org>/<repo>.git
```

- **Watch out:** Check the default branch name (main, master, develop).

Step 8. Build and run locally

Follow the dev-setup guide. Run the app and the tests before changing anything.

```
(project-specific, e.g.) npm install && npm test
```

- **AI:** Paste setup errors into AI and ask for the cause.
- **Watch out:** If setup fails, ask in the channel after searching old messages.

Step 9. Create a branch

One branch per issue, named after the issue.

```
git checkout -b fix/1234-short-name
```

- **Watch out:** Never work directly on main.

Step 10. Understand the code with AI

Open the repo in an AI editor (Antigravity, VS Code + Copilot, Claude Code, Cursor, Zed). Ask it to map the code and trace the bug. Verify with `git log` / `git blame`.

```
git log -p -- path/to/file
git blame path/to/file
```

- **AI:** 'Map this repo; where is <feature> handled? Cite file paths.' 'Trace what happens when <action>.' 'Which existing test is closest to what I need?'
- **Watch out:** AI can be wrong. Check every claim in the actual code.

Step 11. Make the change

Write the fix yourself or with AI help, keeping the change small. Add or update tests.

- **AI:** Use AI for boilerplate, tests and debugging; understand every line.
- **Watch out:** You must be able to explain every line without the AI open.

Step 12. Test and lint

Run the project's tests, linters and formatters exactly like CI does.

```
npm run lint && npm test    (or: make test / go test ./... / pytest)
```

- **AI:** Ask AI to explain failing tests.
- **Watch out:** Never delete or weaken tests to make them pass.

Step 13. Commit

Follow the repo's commit style. Add DCO sign-off if required. Disclose AI use in the format the project asks.

```
git add -p  
git commit -s -m "fix(ui): handle empty trace list (#1234)"
```

- **AI:** AI may suggest a message; rewrite it in your words.
- **Watch out:** AI must never add Signed-off-by; only you can sign the DCO.

Step 14. Sync with upstream

Rebase on the latest upstream branch before pushing.

```
git fetch upstream  
git rebase upstream/main
```

- **Watch out:** Resolve conflicts carefully; re-run tests after rebasing.

Step 15. Push

Push the branch to your fork.

```
git push origin fix/1234-short-name
```

- **Watch out:** Don't force-push after review starts unless the project asks.

Step 16. Open the PR

Fill the PR template: what, why, how tested, screenshots for UI, 'Fixes #1234', AI disclosure.

```
(GitHub UI) Compare & pull request
```

- **AI:** Write the description yourself.
- **Watch out:** Many orgs ban AI-written PR descriptions.

Step 17. Pass CI checks

Watch GitHub Actions. Fix failing lint/test/DCO/CLA checks.

```
Re-run locally, fix, then: git commit -s --amend / git push --force-with-lease
```

- **AI:** Ask AI to explain a CI log.
- **Watch out:** Don't ask maintainers to review a red PR.

Step 18. Respond to review

Reply to every comment yourself, make the changes, and say what you changed.

- **AI:** None for replies (banned in many orgs).
- **Watch out:** Be patient; reviews can take days. Ping politely after a week.

Step 19. Merged! Repeat

After merge, pick a slightly harder issue in the same org. 3-5 merged PRs in one org is a strong signal.

```
git checkout main && git pull upstream main
```

- **Watch out:** Consistency over months matters more than PR count.

Step 20. Apply to GSoC/LFX

Discuss your idea with mentors, then write the proposal yourself: problem, plan, timeline, your merged PRs.

- **AI:** AI only for grammar where the org allows it.
- **Watch out:** AI-written proposals can be rejected automatically.

A worked example

Say you picked a React bug in Jaeger UI: 'Arrow keys in the trace selector resize the minimap'. The full flow looks like this:

```
# 1. comment on the issue and wait to be assigned
# 2. fork jaegertracing/jaeger-ui on GitHub, then:
git clone git@github.com:<you>/jaeger-ui.git
cd jaeger-ui
git remote add upstream https://github.com/jaegertracing/jaeger-ui.git
npm ci          # install exactly what the lockfile says
npm test        # make sure tests pass before you change anything
git checkout -b fix/4178-arrow-keys
# 3. find the code (search, or ask your AI editor to locate the key handler)
# 4. fix it, add a test, run lint + tests again
npm run lint && npm test
git add -p
git commit -s -m "fix(trace): stop arrow keys from resizing minimap (#4178)"
git fetch upstream && git rebase upstream/main
git push origin fix/4178-arrow-keys
# 5. open the PR, fill the template, tick the AI checkbox honestly
```

CHAPTER 8

Using AI tools responsibly

AI editors like Antigravity, Copilot, Claude Code, Cursor and Zed can make you faster. They can also get you banned from a project if you use them carelessly.

Maintainers in 2026 are overwhelmed by low-quality AI-generated PRs and bug reports. curl ended its bug bounty because fewer than 1 in 20 reports were real. Hacktoberfest stopped counting PRs. Many projects now limit how many PRs a newcomer can open. The contributors who stand out are the ones who use AI to learn faster, not to skip understanding.

The golden rule

You own every line you submit

- If you cannot explain a change without the AI open, don't submit it.
- If the project bans AI-generated code, don't use AI to write code there.
- Write your own issue comments, PR descriptions, review replies and proposals.
- Disclose AI use exactly the way the project asks.

AI coding tools in October 2026

Tool	What it is	Free / student access	Link
Google Antigravity	Google's agent-first IDE (editor + agent manager, can control a browser). Launched Nov 2025.	Free 'For Individuals' tier with weekly rate limits; Gemini and Claude models listed on the pricing page. Paid Google AI plans get higher limits.	antigravity.google
GitHub Copilot (VS Code)	Completions, chat and agent mode inside VS Code and other editors.	Copilot Free for everyone. Verified students get the free 'Copilot Student' plan via GitHub Education (replaced free Copilot Pro in Mar 2026).	education.github.com/pack
VS Code agent mode	Built-in agent that edits files and runs terminal commands.	Works with Copilot Free; can also use your own API keys or local models (Ollama).	code.visualstudio.com/blog/2025/04/07/agentMode
Claude Code	Anthropic's agentic coding tool (terminal, IDE, desktop, web).	Needs a paid Claude plan (Pro or higher) or API billing; not in the Free plan. Some universities provide Claude for Education.	code.claude.com/docs/en/overview
Cursor	AI-first code editor (VS Code fork) with agents.	Free Hobby plan with limited agent requests; Pro is paid. The free student year is no longer offered.	cursor.com/pricing
Zed	Fast open-source editor with AI agents.	Free personal plan; 'Zed for Students' gives 12 months of Pro free (university students, 18+).	zed.dev/education
OpenAI Codex CLI	Open-source terminal coding agent.	Sign in with ChatGPT (Free plan included for quick tasks per OpenAI docs) or API key.	developers.openai.com/codex/cli
Gemini CLI	Open-source terminal agent from Google.	Free Google-login tier replaced by Antigravity CLI on 18 Jun 2026; still works with a Gemini API key.	github.com/google-gemini/gemini-cli
Devin Desktop (formerly Windsurf)	AI IDE from Cognition.	Free plan with a light agent quota; Pro is paid.	devin.ai/desktop
Cline / Kilo Code / Aider	Open-source agents for VS Code or terminal.	Free; bring your own API key or use local/open models.	github.com/cline/cline

Free tiers change often. Several changed in mid-2026 (Gemini Code Assist free tier ended, Roo Code shut down). Don't build your whole workflow on one free tool.

Where AI helps most

Task	Good use of AI	Bad use of AI
Understanding the repo	Ask it to map folders, find where a feature lives, trace a call path	Trusting its answer without opening the files
Setup errors	Paste the error, ask for likely causes	Running commands you don't understand
Writing code	Boilerplate, small helpers, suggestions you then review	Generating a whole feature and submitting it
Tests	Suggest edge cases; draft a test you then fix	Changing tests so they pass
Review your own diff	Ask it to list bugs and style problems	Asking an AI bot to review upstream PRs
Communication	Grammar check only, if the org allows	AI-written comments, PR descriptions, proposals

Prompts that help you understand a codebase

Map this repository: what does each top-level folder do, and where is the entry point for <feature>? Cite file paths. Say 'unknown' instead of guessing.

Trace what happens when a user <does X>. List the functions in call order with file:line.

Here is issue #N (pasted). Which files and tests are probably relevant, and why? Don't write code yet.

Show me an existing test similar to what I need and explain how to run just that test.

Summarise CONTRIBUTING.md and the PR template as a checklist for me.

Review my diff for bugs, missing edge cases and style problems against this project's conventions. List problems; don't rewrite it.

Quiz me with 5 questions on how <module> works, then tell me which answers were wrong.

How to disclose AI use

Project style	What to write
Kubernetes	A sentence in the PR description, e.g. 'Parts of this PR were written with the help of generative AI (tool name).' No AI co-author or Assisted-by trailers.
Linux kernel, LLVM, OpenTelemetry, Fedora	An 'Assisted-by:' line in the commit message. Only a human adds Signed-off-by.
Apache projects	A 'Generated-by: ' token in the commit.
Jaeger, matplotlib, others	Tick the AI checkbox / fill the AI section in the PR template.
No rule stated	Ask in the chat. A short honest note in the PR description is a safe default.

AI policies of programs and projects

Program / project	Rule summary
GSoC 2026	Some orgs ban AI entirely, including in proposals. AI only for research/learning/boilerplate/tests; you keep 100% responsibility. AI-written proposals may be auto-rejected.
Outreachy	No generative AI in application essays. Unedited AI contributions you cannot explain = not accepted.
Hacktoberfest 2026	PRs no longer count for rewards because low-effort spam PRs became too easy.

Program / project	Rule summary
Linux Foundation	AI output allowed if the tool's terms and any third-party content are compatible with the license. Projects may be stricter.
CNCF (guidance)	'Own your commit'. Disclosure builds trust; unreviewed AI output is not welcome.
Kubernetes	Disclose AI use in the PR description. No AI co-authors or Assisted-by trailers. No large AI PRs; no AI replies to reviewers.
Linux kernel	AI must never add Signed-off-by. Use an 'Assisted-by:' tag; the human takes full responsibility.
Apache Software Foundation	Allowed with license checks; recommends a 'Generated-by:' token in commits.
Jaeger	You own everything you submit. PR checkbox for AI. Agents must not answer reviews. First-timers limited to 1 open PR.
Jenkins (GSoC)	No copy-pasted AI code, no AI-written PR descriptions or review replies. Two strikes and you are banned.
Zulip	Never answer 'the AI did it'. No AI-generated messages in the community chat.
Oppia (GSoC 2026)	AI allowed, but you are responsible; AI-sounding comments are called out.
NumPy	AI disclosure mandatory; no AI slop; no autonomous agents; do not let AI speak for you.
scikit-learn	No fully automated PRs/issues. Do not paste AI text into issues or PR descriptions.
Django	Disclose the tool and what it did. Run the full test suite. Do not request AI reviews on Django PRs.
CPython	Explain changes in your own words. Never change tests just to make them pass.
OpenTelemetry	Human stays in control; disclose major AI use (Assisted-by: trailer preferred).
LLVM	Human in the loop; no AI on 'good first issue' tasks; no autonomous agents.
Mozilla / Firefox	Responsibility stays with the human; newcomers should not solve beginner bugs purely with AI.
QEMU	Declines contributions believed to include AI-generated content.
Gentoo / NetBSD	AI-generated contributions forbidden (Gentoo) or presumed tainted (NetBSD).
GCC	Declines legally significant LLM-generated contributions (Jul 2026).
Godot	No autonomous agents or vibe coding (auto-ban); disclosure required.
curl	Must reveal if AI found an issue; fake reports = immediate ban. Bug bounty ended Jan 2026 due to AI slop.
Tracker of 200+ project policies	Community list grouping projects by AI rules.

Checklist before you press 'Create pull request'

1. I read the project's AI rules (CONTRIBUTING, AI_POLICY.md, PR template).
2. I can explain every changed line without AI.
3. I ran the full test suite and linters locally, the same way CI does.
4. I did not weaken or delete any test.
5. My PR is small and focused on one issue.
6. I wrote the PR description myself and disclosed AI use as required.
7. No AI added Signed-off-by or a co-author line for me.
8. I did not paste secrets, tokens or private data into an AI tool.

CHAPTER 9

Passing CI checks

Before a maintainer looks at your PR, robots check it. A red check usually means a quick fix you can do yourself.

The common checks

Check	What it wants	How to fix
DCO	Every commit has 'Signed-off-by: Your Name ' matching your Git identity.	<code>git commit --amend -s</code> (last commit) or <code>git rebase --signoff HEAD~3</code> (last 3), then <code>git push --force-with-lease</code>
CLA	You signed the project's Contributor License Agreement (e.g. CNCF EasyCLA for Kubernetes/Headlamp).	Click the CLA bot's link on your PR and sign once with the same email as your commits.
Commit / PR title format	Conventional commits like 'fix(ui): ...' or a prefix like 'frontend: ...'.	Edit the PR title; amend or reword commits (<code>git rebase -i</code>).
Lint / format	Code style rules (ESLint, Prettier, gofmt, golangci-lint, ruff, black).	Run the project's lint/format command locally and commit the result.
Unit tests	All tests pass on every supported version.	Run the failing test locally; read the CI log from the first error.
Build	The project compiles or bundles.	Run the build command locally; check you didn't miss a file.
Branch up to date	Your branch is based on the latest main.	<code>git fetch upstream</code> && <code>git rebase upstream/main</code>
PR template / labels	Filled template, linked issue, sometimes a label or changelog entry.	Edit the PR description; ask a maintainer to add labels.

Commit message formats

```
Conventional commits:
fix(trace): stop arrow keys from resizing minimap (#4178)
feat(ui): add dark-mode toggle to settings
docs: fix broken link in CONTRIBUTING

With DCO sign-off (git commit -s adds the last line):
fix: handle empty list in parser

Signed-off-by: Your Name <you@example.com>
```

Reading a failing check

1. Click 'Details' next to the red check.
2. Scroll to the first error, not the last one.
3. Reproduce it locally with the same command.
4. Fix, commit, push. Don't open a new PR; push to the same branch.
5. If the failure is unrelated to your change (a flaky test), say so politely in a comment.

CHAPTER 10

Communicating with maintainers

How you talk matters as much as how you code. Short, specific, polite messages get answers.

Rules of thumb

- Search the docs and old messages before asking.
- Ask in public channels, not private DMs to maintainers.
- Give context: what you tried, the exact error, your OS and versions.
- One clear question per message.
- Don't ask 'Can I work on this?' on an issue already assigned. Don't comment just '+1' or 'Please assign me'.
- Wait at least a week before a polite follow-up. Maintainers have jobs and exams too.
- Write your messages yourself. Many communities ban AI-written chat messages.

Templates

Introducing yourself in the community chat

```
Hi everyone! I'm <name>, a 3rd year CS student from <college>. I work mostly with
React and TypeScript. I've set up <project> locally and I'm looking at issue #1234.
Happy to be here and looking forward to learning from you all.
```

Asking to work on an issue

```
Hi @maintainer, I'd like to work on this.
I can reproduce it on the latest main. I think the cause is in src/components/Foo.tsx,
where the key handler doesn't check for <X>. I plan to fix it there and add a test.
Could you assign it to me?
```

Asking for help when stuck

```
I'm working on #1234. Running `npm test` fails with:
  TypeError: cannot read properties of undefined (reading 'spans')
I tried <A> and <B>. Node 20, Ubuntu 24.04.
Is there a setup step I'm missing, or is this test expected to fail on main?
```

Pull request description

```
## What
Stop arrow keys in the trace selector from resizing the minimap.

## Why
Fixes #4178. Keyboard users couldn't move through the options.

## How I tested
- Added a unit test for the key handler
- Ran npm run lint && npm test
- Checked manually in Chrome and Firefox (screenshots below)

## AI disclosure
I used <tool> to locate the key handler. I wrote and reviewed the fix myself.
```

A polite follow-up (after about a week)

```
Hi @maintainer, just a gentle ping on this PR when you have time.
I've rebased on the latest main and all checks are green. Happy to change anything.
```

Replying to review

```
Thanks for the review! I've moved the check into the helper as you suggested
and added a test for the empty case (commit abc1234). About the naming, I kept
`isSelectable` because it matches the existing `isVisible`. Happy to rename if you prefer.
```

CHAPTER 11

Writing a winning proposal

A GSoC proposal is a project plan written by you, reviewed by mentors, and backed by your contributions.

Before you write

- Contribute to the org first. Link real merged PRs.
- Discuss the idea with mentors in public channels. Ask what they expect.
- Read proposals that were accepted before (many orgs link them).
- Check the org's rules on AI. Some reject AI-written proposals automatically.

Structure that works

Section	What to include
1. About you	Name, contact, time zone, college and year, GitHub, relevant skills. Keep it short.
2. Your contributions	List merged and open PRs in this org with links and one line each.
3. The problem	What the idea is, why it matters to users, and what exists today (with code references).
4. Technical plan	How you will build it: design, files/modules affected, APIs, diagrams, alternatives you considered.
5. Timeline	Week by week: community bonding, coding weeks, midterm deliverable, final deliverable. Include tests and docs.
6. Testing and docs	How you'll test each part; what docs you'll update.
7. Availability	Hours per week, exam dates, any other commitments. Be honest.
8. After GSoC	How you'll keep maintaining or contributing.

Sample timeline (medium, ~175 hours)

Weeks	Work	Deliverable
Bonding	Finalise design with mentors; set up dev environment; read related code	Approved design doc
1-2	Data model and backend API	PR 1 merged with tests
3-4	UI components	PR 2 with screenshots
5-6	Integration and edge cases	Working feature behind a flag (midterm)
7-8	Tests, performance, accessibility	Test coverage for all new code
9-10	Docs, polish, review feedback	User docs + final PR
11-12	Buffer, final report	Final blog post / report

LFX cover letter

LFX asks for a resume and a short cover letter, not a long proposal. In about one page: why this specific project, what you have already done in it (with PR links), which skills match the project's requirements, your understanding of the task, and your availability for the term.

Common reasons proposals fail

- No contributions to the org, or only trivial ones.
- Copy-paste of the idea description with no technical detail.
- An unrealistic timeline.
- Obvious AI-written text.
- Never talked to mentors before applying.
- Applying to 3 orgs with the same generic proposal.

CHAPTER 12

Beyond code

Some of the most valued contributors rarely write big features. These paths build trust too.

Contribution	How to start	Why it matters
Documentation	Follow the setup guide exactly; fix every step that was unclear or broken	Every newcomer after you benefits
Tests	Find untested helpers; add unit tests	Maintainers love coverage; teaches you the code
Bug triage	Reproduce open bugs on latest main; add steps, versions, logs	Saves maintainers hours
Reviewing	Test other contributors' PRs locally; comment on what you found	Shows judgment; required for Kubernetes membership
Design / UX	Improve UI mockups, accessibility, icons	Many projects lack designers
Translation	Join the project's translation platform	Makes software usable in Indian languages
Community	Answer newcomer questions; write blog posts; give a talk at a FOSS United meetup	Visibility and leadership

CHAPTER 13

Your 12-month plan

October 2026 to September 2027, aimed at LFX 2027 terms and GSoC 2027. 2027 dates are expected, based on 2026.

Month	Your goal	Programs and events
Oct 2026	Learn Git/GitHub; shortlist 2-3 orgs; join their chats; first small PR (docs/typo/test).	Hacktoberfest (events, not PRs). SWoC registration open till Dec 31. SciPy India CFP Oct 19.
Nov 2026	2-3 merged PRs in one main org. Attend a community call.	KubeCon NA talks online. KWOC registration expected late Nov.
Dec 2026	Take a medium issue. Read past GSoC/LFX project ideas of your org.	24 Pull Requests (Dec 1-24). KWOC. Season of KDE applications expected. SWoC closes Dec 31.
Jan 2027	Keep contributing weekly. Watch for LFX Term 1 project listings.	SWoC coding starts Jan 1. LFX T1 applications expected ~late Jan-mid Feb. FOSDEM Jan 30-31 (livestream). Summer of Bitcoin applications expected.
Feb 2027	Study GSoC ideas as soon as orgs are announced; talk to mentors early.	GSoC orgs expected ~late Feb. Outreachy May cohort initial applications expected. ESoC opens (expected).
Mar 2027	Write your GSoC proposal yourself; get mentor feedback.	GSoC contributor applications expected ~mid-late Mar. LFX T1 runs Mar-May. FOSS Hack (expected).
Apr 2027	Keep contributing while results are pending.	C4GT DMP applications expected Apr-May. LFX T2 listings expected mid-Apr. LFDT and OMP applications.
May 2027	If selected: community bonding. If not: apply to LFX T2.	GSoC results expected ~end Apr/early May. LFX T2 applications expected.
Jun-Aug 2027	Coding period, or keep contributing and prepare for LFX T3.	LFX T3 applications expected early Aug. OSPP.
Sep 2027	Share your work (blog/talk). Help newcomers.	LFX T3 starts. IndiaFOSS expected. Hacktoberfest again in Oct.

A weekly routine that works with college

Time per week	Activity
2 hours	Read code and docs of your org; follow recent merged PRs
4-5 hours	Work on your current issue or PR
1 hour	Community: chat, community call, help a newcomer
1 hour	Learn one skill gap (Go, testing, Kubernetes, etc.)

That's about 8-10 hours a week. During exams, reduce it but don't disappear: tell your mentor or the issue thread if you'll be slow.

CHAPTER 14

Events calendar

Contribution drives, conferences and deadlines from October 2026 to November 2027. Many conferences have student tickets, scholarships or free livestreams.

Dates	Event / location	Type	Why go	Link
All of October 2026	Hacktoberfest 2026 Online + 300+ in-person Fests	Contribution drive	PRs no longer count; earn stickers via Fests, livestreams, DEV challenges. Free.	hacktoberfest.com
Oct 1-Dec 31, 2026	Google DevFest 2026 season 800+ events worldwide	Community days	Cheap/free local conferences. India dates below.	developers.google.com
Oct 5-7, 2026	Linux Plumbers Conference 2026 Prague + remote	Conference	Kernel and systems talks; remote attendance possible.	lpc.events
Oct 7-8, 2026	Open Source India (OSI Days) 2026 NIMHANS, Bengaluru	Conference	Subsidised student pass; 6 tracks.	opensourceindia.in
Oct 7-9, 2026	Open Source Summit Europe 2026 Prague	Conference	Watch recordings; scholarships closed.	events.linuxfoundation.org
Oct 9-15, 2026	MLH Global Hack Week: Hacktoberfest Online	Online hack week	Free; counts toward Hacktoberfest stickers.	ghw.mlh.com
Oct 10, 2026	DevFest Noida 2026 Greater Noida	Community day	Local, low cost.	gdg.community.dev
Oct 17, 2026	DevFest Chennai 2026 IIT Madras Research Park, Chennai	Community day	1,200+ attendees.	devfest.gdgchennai.in
Oct 19, 2026 (deadline)	SciPy India 2026 CFP closes	CFP deadline	Submit a talk/poster/workshop for Dec 19-20 at IIT Madras.	scipy.in
Oct 19-20, 2026	All Things Open 2026 Raleigh, USA	Conference	Student scholarship form; free co-located events.	2026.allthingsopen.org
Oct 24, 2026	DevFest Hyderabad 2026 JRC Convention, Hyderabad	Community day	3,000+ attendees.	devfest.gdghyd.in
Oct 25, 2026	DevFest Baroda 2026 Vadodara	Community day	Registration on KonfHub.	gdg.community.dev
Oct 28-29, 2026	GitHub Universe 2026 San Francisco + free virtual pass	Conference	Free virtual pass with live keynotes.	githubuniverse.com
Oct 28-30, 2026	MozFest 2026 Barcelona	Festival	Tech and society.	mozillafoundation.org
Oct 29-31, 2026	React India 2026 Goa	Conference	Scholarships and volunteer roles; announced as the final edition.	reactindia.io
Oct 31, 2026	MumbaiFOSS 2026 IIT Bombay, Powai	Community conference	Free/low cost; good place for a first talk.	fossunited.org
Ongoing (monthly)	FOSS United city meetups Many Indian cities	Meetup series	Free; student speakers welcome.	fossunited.org
Nov 1, 2026	DevFest Gandhinagar 2026 Gandhinagar	Community day	Local.	devfest.gdggandhinagar.dev
Nov 6-12, 2026	MLH Global Hack Week: Builders Week Online	Online hack week	Free, beginner-friendly.	ghw.mlh.com
Nov 9-12, 2026	KubeCon + CloudNativeCon North America 2026 Salt Lake City, USA	Conference	Dan Kohn scholarship deadline Oct 4, 2026. Talks on CNCF YouTube within ~2 weeks.	events.linuxfoundation.org
Nov 18, 2026	gRPCConf India Bengaluru	Conference	India-based CNCF project conference.	events.linuxfoundation.org

Dates	Event / location	Type	Why go	Link
Nov 22, 2026	GopherCon India 2026 Hyderabad	Conference	Go community; CFP on their site.	gopherconindia.org
~late Nov 2026 (expected)	KWoC 2026 registration Online	Contribution drive	Not announced yet; expected from past years.	kwoc.kossiitkgp.org
Nov 30, 2026 (deadline)	FOSSASIA Summit 2027 CFP closes	CFP deadline	Summit is Mar 7-9, 2027 in Bangkok.	eventyay.com
Dec 1-24, 2026	24 Pull Requests Online	Contribution drive	Send PRs through December.	24pullrequests.com
Dec 4-10, 2026	MLH Global Hack Week: Open Source Week Online	Online hack week	Free.	ghw.mlh.com
Dec 5, 2026	OpenSourceCon India 2026 Kolkata	Conference	Bengal's first dedicated OSS conference.	opensourcecon.in
Dec 5, 2026	DevFest Ahmedabad 2026 Ahmedabad	Community day	1,500+ attendees.	devfest26.gdgahmedabad.com
Dec 5, 2026	DevFest Mumbai 2026 Mumbai	Community day	Speaker CFP on Sessionize.	devfestmumbai.com
~mid-Dec 2026 (expected)	Season of KDE 2027 applications Online	Mentorship	Expected from past years.	mentorship.kde.org
Dec 19-20, 2026	SciPy India 2026 IIT Madras, Chennai	Conference	Scientific Python in India.	scipy.in
Dec 31, 2026 (deadline)	SWoC Season 7 registration closes Online	Contribution drive	Coding runs Jan 1-Mar 30, 2027.	swoc.in
Jan 16-17, 2027	MangaloreFOSS 2.0 NIT Karnataka, Surathkal	Community conference	Student-friendly.	fossunited.org
~mid-Jan to mid-Feb 2027 (expected)	LFX Mentorship 2027 Term 1 applications Online	Mentorship	Expected from 2026 dates.	mentorship.lfx.linuxfoundation.org
Jan 30-31, 2027	FOSDEM 2027 Brussels	Conference	Free, no registration. Talks are livestreamed.	fosdem.org
~early-mid Feb 2027 (expected)	Outreachy May 2027 initial applications Online	Internship	Indian university students can apply to this cohort.	outreachy.org
Feb 17-21, 2027	DjangoCon Europe 2027 Innsbruck, Austria	Conference	CFP closes Nov 15, 2026; student tickets.	2027.djangocon.eu
~late Feb 2027 (expected)	GSoC 2027 organizations announced Online	Mentorship	Expected from 2026 dates (Feb 19).	summerofcode.withgoogle.com
Feb 2027 (date TBA)	KCD Pune 2027 Pune	Community day	Local cloud-native event.	cnf.io
Mar 1-May 31, 2027 (expected)	LFX Mentorship 2027 Term 1 runs Online	Mentorship	Expected from 2026 pattern.	docs.linuxfoundation.org
Mar 2-5, 2027	PGConf India 2027 Bengaluru	Conference	CFP closes Oct 15, 2026.	pgconf.in
Mar 7-9, 2027	FOSSASIA Summit 2027 Bangkok	Conference	Student tickets; free virtual passes.	eventyay.com
Mar 15-18, 2027	KubeCon + CloudNativeCon Europe 2027 Barcelona	Conference	CFP closes Oct 11, 2026.	events.linuxfoundation.org
~mid-late Mar 2027 (expected)	GSoC 2027 contributor applications Online	Mentorship	Expected from 2026 dates (Mar 16-31).	summerofcode.withgoogle.com
~Mar 2027 (expected)	FOSS Hack 2027 Online + Indian cities	Hackathon	2026 edition ran in March.	fossunited.org

Dates	Event / location	Type	Why go	Link
Apr 2027 (date TBA)	KCD Hyderabad 2027 Hyderabad	Community day	Local cloud-native event.	cncf.io
~Apr-May 2027 (expected)	C4GT DMP 2027 applications Online	Mentorship	Rs 1 lakh stipend.	codeforgoodtech.in
May 17-19, 2027	Open Source Summit North America 2027 Vancouver	Conference	LF scholarships usually offered.	events.linuxfoundation.org
~May 2027 (expected)	Google I/O 2027 Mountain View + online	Conference	Free livestream.	io.google
~Jun 2027 (expected)	KubeCon + CloudNativeCon India 2027 / Open Source Summit India 2027 India (2026 was Mumbai)	Conference	2026 editions were Jun 16-19 in Mumbai.	events.linuxfoundation.org
Sep 20-22, 2027	Open Source Summit Europe 2027 Glasgow	Conference	LF scholarships.	events.linuxfoundation.org
~Sep 2027 (expected)	IndiaFOSS 2027 Bengaluru (2026)	Conference	Student tickets, diversity scholarship, volunteers.	fossunited.org
October 2027 (expected)	Hacktoberfest 2027 Online	Contribution drive	Every October.	hacktoberfest.com
Nov 8-11, 2027	KubeCon + CloudNativeCon North America 2027 New Orleans	Conference	Dan Kohn scholarships usually offered.	cncf.io

Rows in pink are deadlines. Confirm dates on the official link before you plan travel.

CHAPTER 15

Directory of 100 organisations

Sorted by tier. Activity = commits on the main branch, 1 Sep - 1 Oct 2026. Full links are in the '100 Orgs' sheet.

#	Organisation	Tier	Programs 2026	Stack	Starter repo	Commits
1	Oppia Foundation	Beginner	GSoC 2026	Angular, TypeScript, Python	oppia/oppia	61
2	Zulip	Beginner	GSoC 2026	Python/Django, TypeScript	zulip/zulip	273
3	Meshery	Beginner	GSoC 2026 + LFX 2026 (T1, T2, T3)	React/Next.js, Go	meshery/meshery	317
4	Jaeger (CNCF)	Beginner	GSoC 2026 (via CNCF) + LFX 2026 (T1, T2, T3)	React, TypeScript	jaegertracing/jaeger-ui	73
5	Open Healthcare Network	Beginner	GSoC 2026	React, TypeScript	ohcnetwork/care_fe	38
6	CircuitVerse	Beginner	GSoC 2026	Vue, TypeScript (Rails backend)	CircuitVerse/cv-frontend-vue	25
7	API Dash	Beginner	GSoC 2026	Flutter, Dart	foss42/apidash	33
8	Kiwix	Beginner	GSoC 2026	JavaScript	kiwix/kiwix-js	9
9	OpenMRS	Beginner	GSoC 2026	React, TypeScript	openmrs/openmrs-esm-patient-chart	102
10	Internet Archive	Beginner	GSoC 2026	Python, JavaScript	internetarchive/openlibrary	145
11	Wagtail	Beginner	GSoC 2026	Python/Django, JavaScript	wagtail/wagtail	44
12	Sugar Labs	Beginner	GSoC 2026	JavaScript	sugarlabs/musicblocks	447
13	Learning Equality	Beginner	GSoC 2026	Python/Django, Vue	learningequality/kolibri	241
14	Processing Foundation	Beginner	GSoC 2026	JavaScript	processing/p5.js	169
15	OWASP Foundation	Beginner	GSoC 2026	TypeScript, Node.js, Angular	juice-shop/juice-shop	133
16	JSON Schema	Beginner	GSoC 2026	Next.js, TypeScript	json-schema-org/website	9
17	Accord Project	Beginner	GSoC 2026	React, TypeScript	accordproject/template-playground	7
18	Rocket.Chat	Beginner	GSoC 2026	TypeScript, React, Node.js	RocketChat/Rocket.Chat	180
19	Jitsi	Beginner	GSoC 2026	React, JavaScript	jitsi/jitsi-meet	126
20	Submittity	Beginner	GSoC 2026	PHP, Python, TypeScript	Submittity/Submittity	93
21	omegaUp	Beginner	GSoC 2026	PHP, TypeScript, Vue	omegaup/omegaup	37
22	AnkiDroid	Beginner	GSoC 2026	Kotlin, Android	ankidroid/Anki-Android	564
23	JabRef	Beginner	GSoC 2026	Java	JabRef/jabref	396
24	MetaBrainz Foundation	Beginner	GSoC 2026	Python, React	metabrainz/listenbrainz-server	180
25	OpenStreetMap	Beginner	GSoC 2026	JavaScript	openstreetmap/iD	79
26	SymPy	Beginner	GSoC 2026	Python	sympy/sympy	104
27	stdlib	Beginner	GSoC 2026	JavaScript, C	stdlib-js/stdlib	589
28	FOSSASIA	Beginner	GSoC 2026	Python/Django, JavaScript	fossasia/eventyay	358

#	Organisation	Tier	Programs 2026	Stack	Starter repo	Commits
29	The Mifos Initiative	Beginner	GSoC 2026	Angular, TypeScript	openMF/web-app	99
30	AOSSIE	Beginner	GSoC 2026	Python, React, Tauri	AOSSIE-Org/PictoPy	44
31	Learning Unlimited	Beginner	GSoC 2026	Python/Django	learning-unlimited/ESP-Website	57
32	freeCodeCamp	Beginner	Practice (not in GSoC/LFX)	TypeScript, React, Node.js	freeCodeCamp/freeCodeCamp	329
33	Hoppscotch	Beginner	Practice (not in GSoC/LFX)	TypeScript, Vue	hoppscotch/hoppscotch	10
34	Headlamp (Kubernetes SIG UI)	Intermediate	LFX 2026 (T3)	React, TypeScript, Go	kubernetes-sigs/headlamp	92
35	OpenEverest (Percona)	Intermediate	LFX 2026 (T2, T3)	TypeScript/React, Go	openeverest/openeverest	95
36	PipeCD	Intermediate	GSoC 2026 (via CNCF) + LFX 2026 (T1, T2, T3)	Go, TypeScript	pipe-cd/pipecd	91
37	Kubescape (ARMO)	Intermediate	LFX 2026 (T2, T3)	Go	kubescape/kubescape	177
38	Kyverno (Nirmata)	Intermediate	LFX 2026 (T1, T2, T3)	Go	kyverno/kyverno	87
39	Kubeflow	Intermediate	GSoC 2026 + LFX 2026 (T3)	Python, Go, TypeScript	kubeflow/pipelines	155
40	KubeEdge	Intermediate	LFX 2026 (T1, T2, T3)	Go	kubedge/kubedge	23
41	Volcano	Intermediate	LFX 2026 (T1, T2, T3)	Go	volcano-sh/kthena	15
42	KubeVela	Intermediate	LFX 2026 (T2)	Go	kubevela/kubevela	22
43	LitmusChaos	Intermediate	LFX 2026 (T1)	Go, React	litmuschaos/litmus	17
44	Harbor	Intermediate	LFX 2026 (T1, T2, T3)	Go, Angular	goharbor/harbor	45
45	Lima	Intermediate	LFX 2026 (T2)	Go	lima-vm/lima	109
46	OpenTelemetry	Intermediate	LFX 2026 (T1, T2, T3)	TypeScript	open-telemetry/opentelemetry-js	73
47	OpenKruise	Intermediate	GSoC 2026 (via CNCF) + LFX 2026 (T1, T2, T3)	Go	openkruise/agents	45
48	KubeArmor	Intermediate	LFX 2026 (T2)	Go, eBPF	kubearmor/KubeArmor	16
49	Kuadrant	Intermediate	LFX 2026 (T2)	Go	Kuadrant/kuadrant-operator	233
50	Microcks	Intermediate	LFX 2026 (T2)	Java, Angular	microcks/microcks	61
51	Apicurio (Red Hat)	Intermediate	LFX 2026 (T3)	Java, React	Apicurio/apicurio-registry	160
52	krkn (Chaos)	Intermediate	LFX 2026 (T1, T2)	Python	krkn-chaos/krkn	20
53	Jenkins	Intermediate	GSoC 2026	Java, JavaScript	jenkinsci/jenkins	98
54	Joplin	Intermediate	GSoC 2026	TypeScript, React Native	laurent22/joplin	181
55	webpack	Intermediate	GSoC 2026	JavaScript	webpack/webpack	374

#	Organisation	Tier	Programs 2026	Stack	Starter repo	Commits
56	Gemini CLI (Google)	Intermediate	GSoC 2026	TypeScript	google-gemini/gemini-cli	60
57	Django Software Foundation	Intermediate	GSoC 2026	Python	django/django	63
58	checkstyle	Intermediate	GSoC 2026	Java	checkstyle/checkstyle	254
59	cBioPortal	Intermediate	GSoC 2026	React, TypeScript	cBioPortal/cbioportal-frontend	18
60	OpenWISP	Intermediate	GSoC 2026	Python/Django	openwisp/openwisp-controller	8
61	Neuroinformatics Unit	Intermediate	GSoC 2026	Python	neuroinformatics-unit/movement	11
62	Kornia	Intermediate	GSoC 2026	Python, PyTorch	kornia/kornia	542
63	OpenVINO Toolkit (Intel)	Intermediate	GSoC 2026	C++, Python	openvinotoolkit/openvino	325
64	Apache Software Foundation	Intermediate	GSoC 2026	Python	apache/airflow	816
65	Metasploit	Intermediate	GSoC 2026	Ruby	rapid7/metasploit-framework	26
66	AboutCode	Intermediate	GSoC 2026	Python	aboutcode-org/scancode-toolkit	17
67	Graphite	Intermediate	GSoC 2026	Rust, Svelte	GraphiteEditor/Graphite	102
68	KDE Community	Intermediate	GSoC 2026	C++, Qt	KDE/kdenlive	98
69	PostgreSQL (pgAdmin)	Intermediate	GSoC 2026	Python, React	pgadmin-org/pgadmin4	200
70	OpenAstronomy	Intermediate	GSoC 2026	Python	sunpy/sunpy	8
71	Wikimedia Foundation	Intermediate	GSoC 2026	PHP, JavaScript	wikimedia/mediawiki	291
72	Open Food Facts	Beginner	GSoC 2026	Flutter, Dart	openfoodfacts/smooth-app	44
73	Kubernetes	Advanced	LFX 2026 (T1, T2, T3)	Go	kubernetes/kubernetes	700
74	Prometheus	Advanced	LFX 2026 (T1)	Go	prometheus/prometheus	221
75	Cilium	Advanced	LFX 2026 (T1)	Go, eBPF, C	cilium/cilium	547
76	etcd	Advanced	LFX 2026 (T1)	Go	etcd-io/etcd	52
77	WasmEdge	Advanced	GSoC 2026 (via CNCF) + LFX 2026 (T1, T2, T3)	C++, Rust	WasmEdge/WasmEdge	66
78	Podman (containers)	Advanced	LFX 2026 (T3)	Go	containers/podman	209
79	KubeVirt	Advanced	GSoC 2026	Go	kubevirt/kubevirt	280
80	Linux Kernel (LKMP)	Advanced	LF Kernel Mentorship (LKMP) + GSoC 2026 (The Linux Foundation)	C	torvalds/linux	1895
81	LLVM	Advanced	GSoC 2026	C++	llvm/llvm-project	4724
82	GCC	Advanced	GSoC 2026	C, C++	gcc-mirror/gcc	971
83	FFmpeg	Advanced	GSoC 2026	C, Assembly	FFmpeg/FFmpeg	657
84	QEMU	Advanced	GSoC 2026	C	qemu/qemu	1002
85	Git	Advanced	GSoC 2026	C	git/git	110
86	Rust Foundation	Advanced	GSoC 2026	Rust	rust-lang/rust	2172

#	Organisation	Tier	Programs 2026	Stack	Starter repo	Commits
87	Blender Foundation	Advanced	GSoC 2026	C++, Python	blender/blender	752
88	LibreOffice	Advanced	GSoC 2026	C++	LibreOffice/core	702
89	The Julia Language	Advanced	GSoC 2026	Julia, C	JuliaLang/julia	346
90	CERN-HSF	Advanced	GSoC 2026	C++	root-project/root	432
91	ArduPilot	Advanced	GSoC 2026	C++	ArduPilot/ardupilot	522
92	FreeCAD	Advanced	GSoC 2026	C++, Python	FreeCAD/FreeCAD	496
93	Neovim	Advanced	GSoC 2026	C, Lua	neovim/neovim	385
94	Ceph	Advanced	GSoC 2026	C++, Python	ceph/ceph	100
95	Open Robotics	Advanced	GSoC 2026	C++	gazebo/gz-sim	22
96	SageMath	Advanced	GSoC 2026	Python, Cython	sagemath/sage	173
97	Mixxx	Advanced	GSoC 2026	C++, Qt	mixxxdj/mixxx	146
98	dora-rs	Advanced	GSoC 2026	Rust	dora-rs/dora	236
99	Swift	Advanced	GSoC 2026	C++, Swift	swiftlang/swift	1294
100	LF Decentralized Trust (Hyperledger)	Advanced	LFDT Mentorship 2026 (LF)	Go	hyperledger/fabric	17

CHAPTER 16

Common mistakes and FAQs

Learn from the mistakes that get newcomers ignored or banned.

Mistakes to avoid

Mistake	Do this instead
Commenting 'please assign me' on many issues	Pick one issue, show you understand it, then ask
Opening a PR for an issue assigned to someone else	Find another issue or ask if it's still active
One huge PR touching 30 files	Small PRs, one concern each
Submitting AI output you don't understand	Understand and test every line; follow the AI policy
Ignoring the PR template and CI failures	Fill the template; get checks green before asking for review
Pinging maintainers daily	Wait a week, then one polite ping
Arguing with review comments	Ask questions, explain your reasoning, accept decisions
Starting only in March before GSoC	Start in October; consistency is visible
Contributing to 10 orgs at once	One main org, one backup
Giving up after one rejected PR	Ask what to improve; try the next issue

Frequently asked questions

I'm in 2nd year and only know HTML/CSS/JS. Is it too early?

No. Start with a Beginner web org (Chapter 5). Documentation and small UI fixes are real contributions, and 2nd year is the best time to start.

How many PRs do I need before applying to GSoC or LFX?

There's no official number. Some orgs set a minimum. What matters is meaningful merged work in the org you apply to and good discussion with mentors.

Can I apply to both GSoC and LFX?

Yes, they're separate programs. Remember that LFX selects you only once in your life, and GSoC at most twice, so choose projects you really want.

Can I use AI tools?

Yes, where the project allows it, and only for work you fully understand. Some projects ban AI code. Proposals and messages should be your own. See Chapter 8.

Is Hacktoberfest still worth it?

Yes for events and learning, but in 2026 PRs don't earn rewards. Use October to start with your chosen org instead.

A maintainer hasn't replied in two weeks. What now?

Post one polite ping. If there's still no reply, ask in the community chat or move to another issue. Don't take it personally.

My PR was closed without merge. Did I fail?

No. Ask what would have made it acceptable and apply that to the next one. Everyone has closed PRs.

Do I need a powerful laptop?

Most web and Python projects run on 8 GB RAM. Kubernetes and big C++ projects need more; use kind with small clusters or cloud IDEs like GitHub Codespaces where allowed.

What if my English isn't great?

Keep sentences short and clear. Maintainers come from all over the world. A grammar checker is fine where the org allows it.

APPENDIX A

Git cheatsheet

Commands you'll use for every contribution. Replace , and .

Task	Command
Clone your fork	<code>git clone git@github.com:<you>/<repo>.git</code>
Add upstream	<code>git remote add upstream https://github.com/<org>/<repo>.git</code>
Check remotes	<code>git remote -v</code>
New branch	<code>git checkout -b fix/123-short-name</code>
See changes	<code>git status / git diff</code>
Stage parts of files	<code>git add -p</code>
Commit with DCO sign-off	<code>git commit -s -m "fix: short description (#123)"</code>
Amend last commit	<code>git commit --amend -s</code>
Update from upstream	<code>git fetch upstream && git rebase upstream/main</code>
Push branch	<code>git push origin fix/123-short-name</code>
Push after rebase	<code>git push --force-with-lease</code>
Squash last 3 commits	<code>git rebase -i HEAD~3 (mark commits as 'squash')</code>
Undo unstaged changes	<code>git restore <file></code>
See history of a file	<code>git log -p -- <file></code>
Who changed a line	<code>git blame <file></code>
Save work temporarily	<code>git stash / git stash pop</code>
Sync your fork's main	<code>git checkout main && git pull upstream main && git push origin main</code>
Delete merged branch	<code>git branch -d fix/123-short-name && git push origin --delete fix/123-short-name</code>

Fixing common Git problems

Problem	Fix
Forgot to sign off commits	<code>git rebase --signoff HEAD~N</code> then <code>git push --force-with-lease</code>
Committed to main by mistake	<code>git branch fix/x, git reset --hard upstream/main, then git checkout fix/x</code>
Merge conflict during rebase	Edit the conflicted files, <code>git add</code> them, <code>git rebase --continue</code>
Wrong email on commits	Set the right email, then <code>git commit --amend --reset-author (last commit)</code>
Too many small commits	<code>git rebase -i upstream/main</code> and squash, if the project asks for it

APPENDIX B

Resources

Including the reference links you shared.

Category	Name	Link	Note
Your sources	Ossium	ossium.in	Discovery platform: GSoC orgs, trending repos, good first issues, bounties. Free, no login.
Your sources	20 Open Source Internship Programs (dev.to)	dev.to/yakhilesh/20-open-source-internship-programs-updated-for-2026-5h32	Program list. Some entries are outdated (see the Programs sheet for 2026 corrections).
Your sources	List-of-OpenSource-Programs (vikas0799)	github.com/vikas0799/List-of-OpenSource-Programs	Programs, competitions and a 2026 timeline.
Your sources	Google Summer of Code	summerofcode.withgoogle.com	Official site: orgs, timeline, FAQ.
Your sources	LFX Mentorship	mentorship.lfx.linuxfoundation.org/#projects_all	Official LFX project listings.
Official	GSoC timeline	developers.google.com/open-source/gsoc/timeline	Dates for the current year.
Official	GSoC AI guidance	developers.google.com/open-source/gsoc/resources/ai_guidance	What AI use is allowed.
Official	CNCF mentoring repo	github.com/cncf/mentoring	All CNCF LFX/GSoC project ideas by term.
Official	LFX stipend amounts	docs.linuxfoundation.org/lfx/mentorship/mentee-stipends	Country-wise stipend.
Official	LFX program timelines	docs.linuxfoundation.org/lfx/mentorship/mentorship-program-timelines	Term dates.
Find issues	Good First Issues	goodfirstissues.com	Curated beginner issues.
Find issues	Up For Grabs	up-for-grabs.net	Projects looking for contributors.
Find issues	First Timers Only	www.firsttimersonly.com	Guide + issues for first-timers.
Find issues	CodeTriage	www.codetriage.com	Get issues in your inbox.
Find issues	Awesome for Beginners	github.com/MunGell/awesome-for-beginners	Beginner-friendly projects by language.
Find issues	GSoC 2026 explorer	github.com/VincenzoImp/gsoc-2026-explorer	All 185 GSoC 2026 orgs and ideas, searchable.
Learn	Pro Git book	git-scm.com/book	Free, complete Git guide.
Learn	Learn Git Branching	learngitbranching.js.org	Interactive visual Git practice.
Learn	GitHub Skills	skills.github.com	Hands-on GitHub courses.
Learn	Open Source Guides	opensource.guide/how-to-contribute	How to contribute, by GitHub.
Learn	LFD103: Beginner's Guide to Linux Kernel Development	training.linuxfoundation.org/training/a-beginners-guide-to-linux-kernel-development-lfd103	Free; required for LKMP.
Learn	Conventional Commits	www.conventionalcommits.org	Commit message format many repos use.
Learn	Developer Certificate of Origin (DCO)	developercertificate.org	What 'git commit -s' means.
AI	Open source AI contribution policies tracker	github.com/melissawm/open-source-ai-contribution-policies	200+ projects grouped by AI rules.

APPENDIX C

Glossary

Words you'll see in every open-source project.

Term	Meaning
Fork	Your own copy of a repository on GitHub.
Upstream	The original repository you forked from.
Pull request (PR)	A request to merge your branch into the project. GitLab calls it a merge request (MR).
Maintainer	Someone with permission to review and merge.
CI	Continuous integration: automated checks run on every PR.
DCO	Developer Certificate of Origin. 'Signed-off-by' line confirming you have the right to submit the code.
CLA	Contributor License Agreement you sign once before contributing.
Rebase	Replaying your commits on top of the latest main.
Squash	Combining several commits into one.
Good first issue	An issue maintainers marked as suitable for newcomers.
Triage	Sorting and reproducing issues.
LGTM	'Looks good to me', an approval.
Nit	A tiny, optional suggestion in review.
Draft PR	A PR that isn't ready for review yet.
Community bonding	GSoC's first month for getting to know mentors and the code.
Mentor	An experienced contributor who guides you during a program.
AGENTS.md	A file with instructions that AI coding agents read automatically.
AI_POLICY.md	A project's rules for contributors who use AI.

Facts in this handbook were checked against official program pages, [cncf/mentoring](#), the GSoC 2026 organisation list and each repository's git history on 1 October 2026. Programs change every year; always confirm on the official link.